

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

MODEL QUESTION PAPER

RAILWAY AND BRIDGE ENGINEERING

Time: 3 hours

Maximum marks: 100 Marks

Part - A

MAXIMUM MARK 100

I Answer **ALL** questions in one or two sentences. Each question carries 2 marks.

1. Differentiate locomotive & coach
2. What is known as sleeper density?
3. Differentiate points and crossing.
4. What is known as vertical clearance?
5. What is approach channel in harbor?

Part – B

MAXIMUM MARK 30

II Answer **ANY FIVE** questions. Each question carries 6 marks.

1. Explain the function and characteristics of ballast
2. What is known as bearing plate .Explain its merits and demerits?
3. Explain with sketch coloured light signal.
- 4 What is known as super elevation? Give its formula
5. Explain how depth of foundation for bridge is determined. Give its formula?
- 6 What are advantages and disadvantages of “through bridge”?
7. What are main factor to be consider for the layout of an air port.

Part – C

MAXIMUM MARK 60

(Answer one full question from each unit. Each question carries 15 marks)

UNIT – I

- | | | |
|------------|---|---|
| III | (a) Explain different types of rails with sketches. | 7 |
| | (b)What are the different types of welding used in railway track? | 8 |

OR

IV	(a) Explain with sketch chairs and keys	7
	(b) Explain the term	
	(i) Corrugated rail (ii) corrosion of rail (iii) kinks of rail	8
UNIT - II		
V	(a) Explain the classification of different types of gradient	8
	(b) Draw typical sketch of split switch (point set for main line).	7
OR		
VI	(a) Explain with sketch the working of semaphore signal	7
	(b) Explain the classification of interlocking with sketch	8
UNIT - III		
VII	(a) Explain various load and forces to be considered for the design of road bridges	6
	(b) Explain with sketch any three types of RCC bridge.	9
OR		
VIII	(a) Explain various classification of abutment with or without wing wall.	7
	(b) Explain with sketch any four types of pier with sketch	8
UNIT - IV		
IX	(a) What is tunnel lining. Mention any five objects.	7
	(b) Explain various classification of harbour.	8
OR		
X	(a) Draw the cross section of tunnel for a single lane broad gauge track marking all its parts	7
	(b) What is an air port? Give sketch of any three runway layout.	8
